Project Module 2:
Explainability System Design

German Credit Dataset

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Part 1: Model Training

1.1 Model Choice

We trained a **Decision Tree** classifier (`max_depth=5`, `criterion="gini"`, `min_samples_leaf=10`, `random_state=42`). This model was selected because it provides a qualitatively different form of interpretability from the Logistic Regression used in Module 1, enabling a richer explainability analysis in Parts 2–6. The same 80/20 stratified train/test split and 65-feature preprocessing pipeline from Module 1 were reused without modification (800 training / 200 test samples).

Hyperparameter rationale. `max_depth=5` caps the longest decision path at five splits, keeping explanations cognitively tractable; an unconstrained tree grows beyond depth 15 and overfits. `min_samples_leaf=10` (1.25% of training data) prevents leaf nodes built on very small sub-groups, reducing variance. `criterion="gini"` is the standard efficient split criterion, equivalent to entropy in practice. The tree reaching exactly depth 5 confirms the cap is genuinely constraining.

1.2 Predictive Performance

1.2.1 *Overall Metrics*

Metric	Decision Tree (Mod. 2)	LR Baseline (Mod. 1)	Diff.
Accuracy	0.7100	0.7050	+0.0050
Precision	0.7253	0.7755	−0.0502
Recall	0.9429	0.8143	+0.1286
F1-score	0.8199	0.7944	+0.0255

Table 1: Overall metrics on the held-out test set (200 samples).

1.2.2 *Per-Class Breakdown*

Class	Precision	Recall	F1	Support
Bad Credit (0)	0.56	0.17	0.26	60
Good Credit (1)	0.73	0.94	0.82	140
Macro avg	0.64	0.55	0.54	200
Weighted avg	0.67	0.71	0.65	200

Table 2: Classification report by class.

1.2.3 *Metric Interpretation*

Accuracy (0.71) is marginally above the baseline but misleading under 70/30 class imbalance; the per-class breakdown is more informative.

Precision (0.7253) is 5 pp below baseline: the tree approves a slightly higher proportion of truly bad applicants, trading precision for recall.

Recall (0.9429) is 12.9 pp above baseline: the model captures 94% of genuinely creditworthy applicants, reducing wrongful rejections.

F1-score (0.8199) exceeds baseline because the large recall gain outweighs the precision loss.

Critical concern — Bad Credit recall (0.17). Only 10 of 60 truly bad applicants are correctly identified; the remaining 50 would receive loans they are unlikely to repay. This

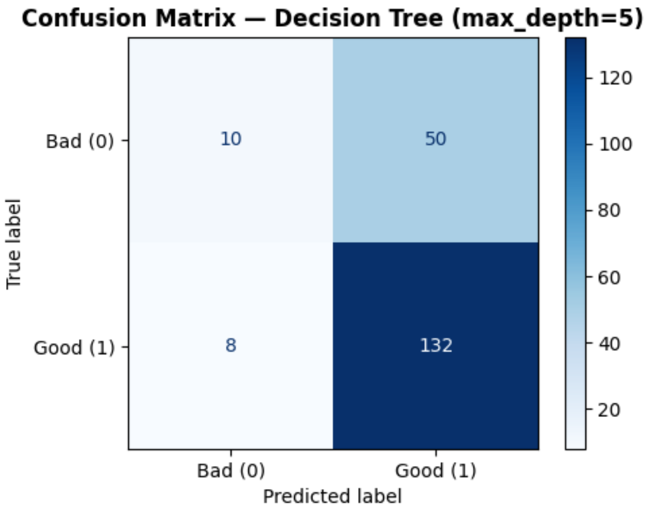

Figure 1: Confusion matrix for the Decision Tree on the held-out test set (200 samples).

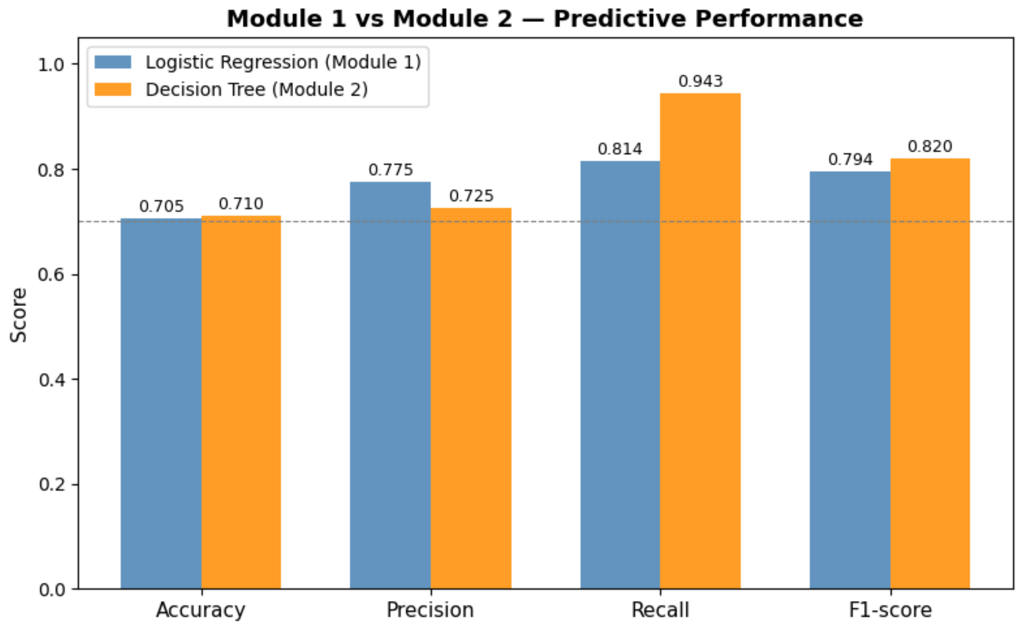

Figure 2: Side-by-side metric comparison: Decision Tree (Module 2) vs. Logistic Regression baseline (Module 1).

is a severe operational flaw: a naïve "approve everyone" classifier would reach 70% accuracy while producing the same catastrophic Bad-class recall. The root cause is the 70/30 class imbalance: without cost-sensitive learning the tree optimises for the majority class. Mitigations include `class_weight="balanced"`, lowering the classification threshold below 0.5, or SMOTE oversampling; all are prerequisites for deployment and are addressed in Part 7.

1.3 Summary

The Decision Tree matches the Module 1 baseline in overall accuracy and F1 while providing richer interpretability. The primary concern is the very low Bad-class recall; the goal of Module 2 is interpretability, not raw performance improvement, so this trade-off is acceptable for the analysis but must be resolved before deployment.

Part 2: Why Is This Model Interpretable?

2.1 Structural Transparency

A Decision Tree encodes its entire decision logic as binary if/then rules that any human can follow step by step. There are no hidden activations or weight matrices; every prediction traces to an explicit sequence of threshold comparisons.

Property	Value
Maximum permitted depth	5
Actual depth reached	5
Total nodes	43
Internal (decision) nodes	21
Leaf (prediction) nodes	22

Table 3: Tree structural statistics (Notebook Cell 10).

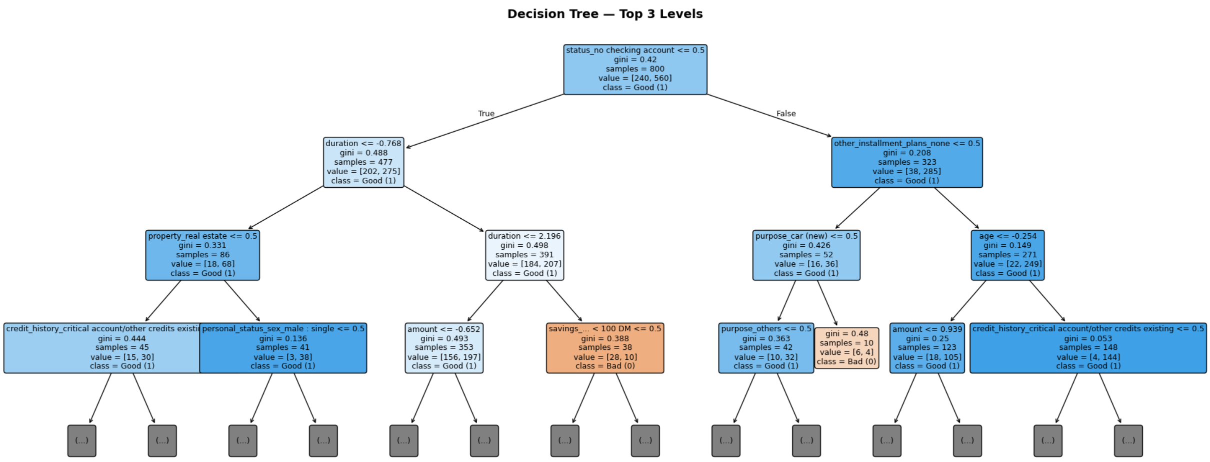

Figure 3: Top-3-level view of the Decision Tree. Each node shows the split feature, threshold, Gini impurity, sample count, and class distribution.

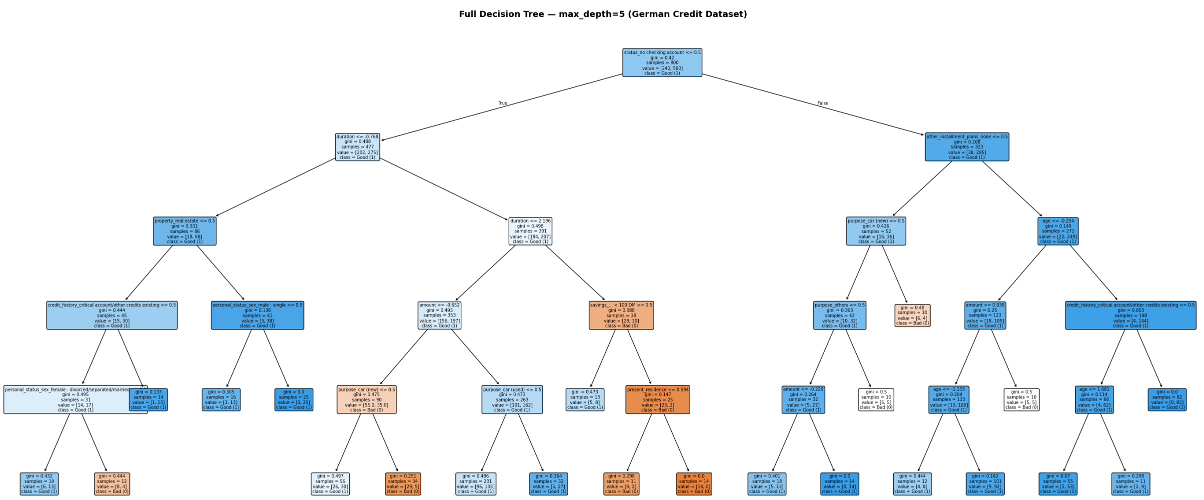

Figure 4: Full Decision Tree at `max_depth=5`. All 43 nodes and 22 leaf profiles are visible.

2.2 Four Properties that Make Predictions Understandable

Property 1 — Global Transparency. The full model can be rendered as a readable flowchart (*decision_tree_full.png*, Notebook Cell 7). A bank officer, regulator, or auditor can inspect all decision logic without any supporting tools—impossible with neural networks and non-trivial with Logistic Regression.

Property 2 — Local Traceability. Any individual prediction is fully explained by the root-to-leaf path matching that applicant's feature values. Each node states a condition and the direction taken, giving a complete causal account of approval or rejection.

Property 3 — Bounded Complexity. `max_depth=5` limits any explanation to at most five yes/no questions, well within the 7±2 cognitive load threshold established in human-factors research. The 22 leaf nodes yield 22 distinct decision profiles—a manageable number for policy documentation.

Property 4 — No Mathematical Black Box. Each node shows the exact feature name, split threshold, Gini impurity, sample count, and class distribution. No post-hoc approximation is needed. Logistic Regression, by contrast, requires interpreting scaled coefficients; ensemble methods require averaging thousands of trees.

2.3 Contrast with Logistic Regression

Logistic Regression models **additive linear contributions** to a log-odds score and cannot natively represent conditional interactions such as *"if checking account is empty AND duration is long AND purpose is used car, then Bad"*. Decision Trees represent precisely these structures, making them the more appropriate choice for the local-explanation and split-level analyses required in Parts 4–6.

Part 3: What Features Matter the Most?

3.1 Method — Gini Feature Importance

Gini feature importance (Mean Decrease in Impurity) is the weighted sum of impurity reductions across all nodes where a feature is used as the split criterion, normalised to sum to 1.0.

Known limitation — bias towards continuous features. Gini importance systematically overestimates continuous numeric features (*duration*, *amount*) relative to binary one-hot indicators, because continuous features offer more candidate split points and therefore more opportunities to reduce impurity. The reported importances for *duration* (0.1585) and *amount* (0.1103) may therefore be inflated. Permutation importance, computed on held-out data, would provide a bias-corrected alternative and is recommended before any policy use of these rankings.

3.2 Top 10 Features

3.3 Feature-by-Feature Analysis

Feature 1 — `status_no checking account` (0.3662). The single most important feature and the **root split** of the tree. Absence of a checking account signals no formal banking relationship and no transaction history—the primary indicator assessed in classical credit scoring.

Feature 2 — `duration` (0.1585). Longer loans expose the lender to extended repayment risk and correlate with financially stretched borrowers; a 48-month loan carries fundamentally different risk to a 6-month one at the same principal.

Feature 3 — `amount` (0.1103). Larger loans directly increase potential loss on default and proxy for the borrower's income level, collateral value, and repayment capacity.

#	Feature	Original Column	Importance	Cum. %
1	`status_no checking account`	status	0.3662	36.6%
2	`duration`	duration	0.1585	52.5%
3	`amount`	amount	0.1103	63.5%
4	`purpose_car (new)`	purpose	0.0865	72.2%
5	`savings_ < 100 DM`	savings	0.0498	77.1%
6	`other_installment_plans_none`	other_installment_plans	0.0455	81.7%
7	`purpose_car (used)`	purpose	0.0384	85.5%
8	`age`	age	0.0364	89.2%
9	`credit_history_critical...`	credit_history	0.0311	92.3%
10	`property_real estate`	property	0.0295	95.2%

Table 4: Top 10 features by Gini importance. Top 10 cover 95.2% of total importance.

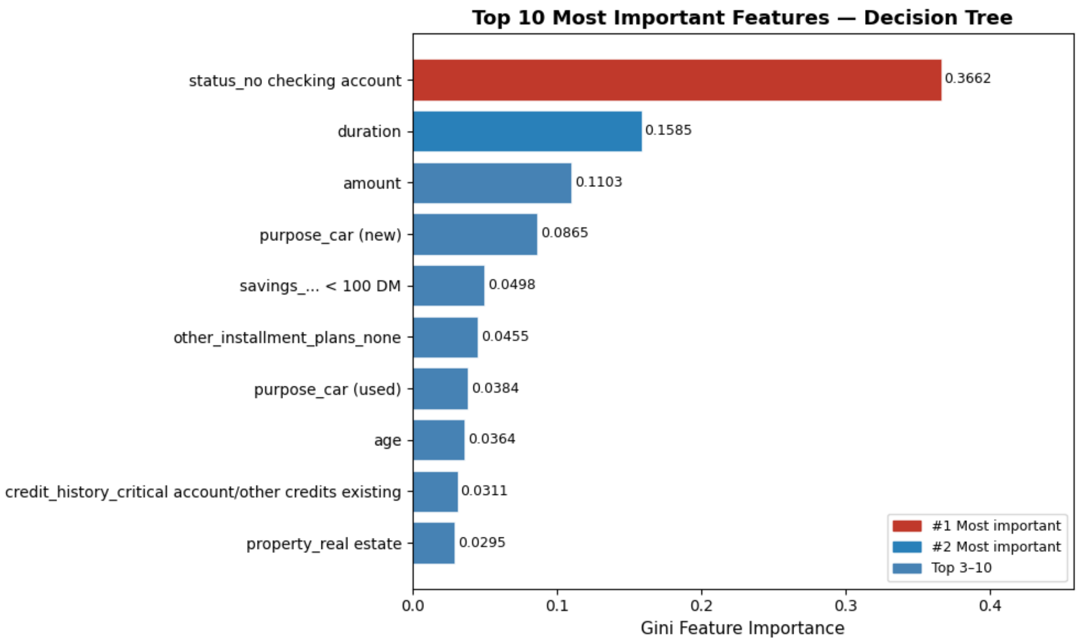

Figure 5: Top 10 most important features by Gini importance (horizontal bar chart).

Feature 4 — `purpose_car (new)` (0.0865). New car purchases are large discretionary expenditures often associated with inflated dealer pricing; buyers may overextend their finances relative to other loan purposes.

Feature 5 — `savings < 100 DM` (0.0498). Minimal savings indicate no financial buffer for missed payments during income disruption, corresponding to the *Capital* component of the 5 C's of Credit.

Feature 6 — `other_installment_plans_none` (0.0455). No other instalments can signal either a clean financial record (positive) or limited credit history (ambiguous); the tree uses this to refine splits after the top-level checking account and duration splits.

Feature 7 — `purpose_car (used)` (0.0384). Used car loans involve older collateral with uncertain valuations. The appearance of both car-purpose features in the top 10 confirms the tree distinguishes between the two categories, which carry different default rates.

Feature 8 — `age` (0.0364). Age correlates legitimately with employment stability and credit history length. However, age is also a **protected attribute** (under_25 vs. age_25_and_over, Module 1). Its direct presence as a split feature may explain the fairness disparities observed in

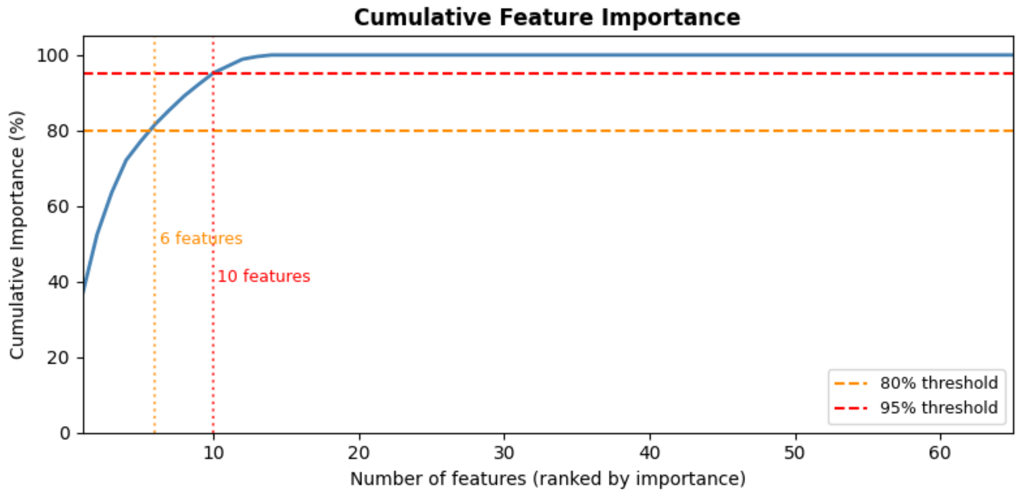

Figure 6: Cumulative feature importance curve. The dashed lines mark the 80% and 95% thresholds.

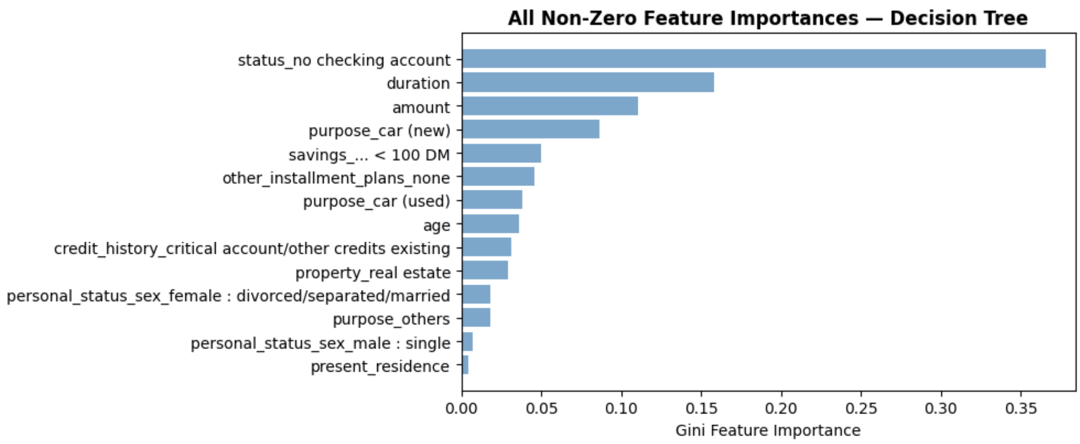

Figure 7: All non-zero feature importances. Confirms that 55 of 65 features contribute negligibly.

Module 1 (under_25 TPR gap: 0.2137; ECE gap: 0.2975) and warrants investigation in Part 6.

Feature 9 — `credit_history_critical...` (0.0311). A critical account history signals past default or delays—the *Character* component of the 5 C's and one of the strongest predictors of future repayment.

Feature 10 — `property_real estate` (0.0295). Real estate ownership is a tangible collateral signal corresponding to the *Collateral* component of the 5 C's of Credit.

3.4 Domain Alignment and Notable Concern

All five components of the **5 C's of Credit** are represented across the top 10 features, indicating the model has re-discovered standard banking logic from data alone. The one notable concern is `age` at rank 8: while partially legitimate, its role as a protected demographic attribute warrants the further investigation conducted in Part 6.

3.5 Concentration of Importance

The distribution is **highly concentrated**: the top feature alone accounts for 36.6%, the top 3 for 63.5%, and the top 5 for 77.1 %.

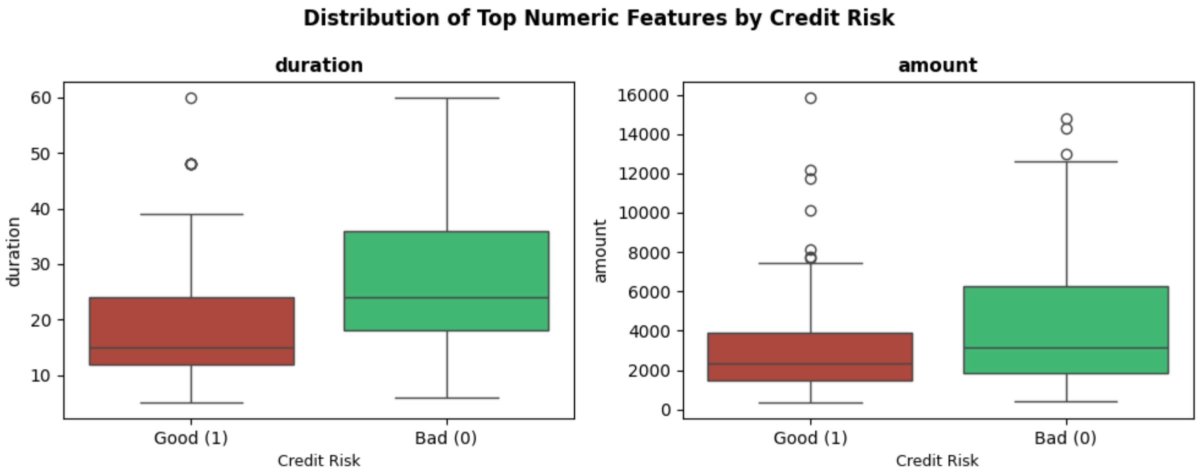

Figure 8: Distribution of top numeric features (*duration*, *amount*) split by credit risk class (Good vs. Bad). The separation between classes confirms their predictive relevance.

- **Positive:** The model is parsimonious—a credit officer needs to internalise only 3–5 rules to understand most decisions, making the system practical to audit.
- **Risk:** Heavy reliance on a single feature (checking account status) makes the model vulnerable if that feature is missing, manipulated, or correlated with protected group membership.

Notebook Cell References

Cell 3	Decision Tree metrics and classification report
Cell 7	Full decision tree plot (*decision_tree_full.png*)
Cell 8	Top-3-level compact tree (*decision_tree_top3.png*)
Cell 10	Tree structural statistics
Cell 11	Top-10 feature importances table and cumulative %
Cell 12	Feature importance bar chart
Cell 13	Cumulative importance curve
Cell 15	Box plots of top numeric features by class

Part 4: How Do Features Affect Predictions?

We look at the **direction** and **behaviour** of the top features for both models.

- **Logistic Regression:** the sign of the coefficient (+ or −) shows whether a feature increases or decreases $P(\text{Good})$.
- **Decision Tree:** the number of splits and their thresholds show whether a feature has a fixed effect or a changing one.

4.1 Logistic Regression — Direction of Effect

A positive coefficient means the feature pushes the prediction toward Good credit. A negative coefficient means it pushes toward Bad credit.

Table 5: Top 10 LR features by absolute coefficient value (Notebook Cell 29).

#	Feature	Coefficient	Direction
1	`purpose_car (used)`	+1.3220	Increases P(Good)
2	`status_no checking account`	+0.9368	Increases P(Good)
3	`other_debtors_guarantor`	+0.9091	Increases P(Good)
4	`credit_history_critical...`	+0.7832	Increases P(Good)
5	`status_...<100 DM`	−0.7523	Decreases P(Good)
6	`purpose_retraining`	−0.7302	Decreases P(Good)
7	`purpose_car (new)`	−0.7125	Decreases P(Good)
8	`foreign_worker_no`	+0.6124	Increases P(Good)
9	`other_debtors_co-applicant`	−0.6091	Decreases P(Good)
10	`foreign_worker_yes`	−0.6064	Decreases P(Good)

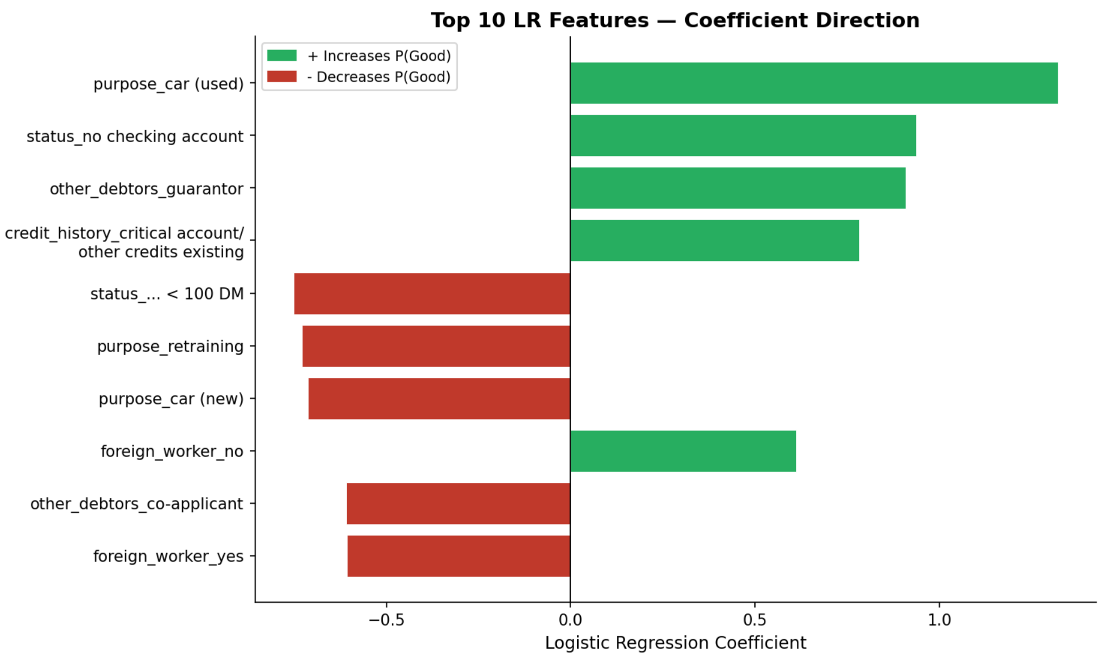

Figure 9: Top 10 LR coefficients. Green bars increase $P(\text{Good})$; red bars decrease it (Notebook Cell 29).

Features that increase P(Good):

- `purpose_car (used)` (+1.3220): the strongest positive feature. Used car loans are smaller and lower risk than new car loans.
- `status_no checking account` (+0.9368): this is also the root split of the Decision Tree. Although having no checking account might seem risky, the model learned that this group tends to repay in this dataset. This is worth looking into in Part 6.
- `other_debtors_guarantor` (+0.9091): having a guarantor gives the bank a backup if the borrower cannot repay, which lowers risk.
- `credit_history_critical...` (+0.7832): this looks counterintuitive since a critical credit history sounds bad. The model may be picking up on applicants who have already cleared past issues. This is flagged for Part 6.
- `foreign_worker_no` (+0.6124): non-foreign workers tend to have longer local work records in this dataset.

Features that decrease P(Good):

- `status_...<100 DM` (−0.7523): a very low checking balance signals financial stress and a higher chance of missing payments.
- `purpose_retraining` (−0.7302): loans for retraining are linked to job loss or career change, which means less stable income.
- `purpose_car (new)` (−0.7125): new cars are expensive and buyers may take on more debt than they can comfortably repay.
- `other_debtors_co-applicant` (−0.6091): having a co-applicant may signal that the main borrower has a weaker individual profile.
- `foreign_worker_yes` (−0.6064): this pushes the prediction toward Bad credit. Since it is linked to nationality, it raises a fairness concern discussed in Part 6.

4.2 Decision Tree — Split Behaviour

A feature used in more than one node applies different thresholds to different groups, so its effect **changes** across values. A feature used in only one node has a **fixed** binary effect.

Table 6: Top 10 DT features by number of splits (Notebook Cell 28). Numeric thresholds are in standardised units; binary features split at 0.5.

#	Feature	Splits	Thresholds	Effect
1	`amount`	3	−0.65, −0.23, +0.94	Changes across loan amount ranges
2	`age`	3	−0.25, −1.13, +1.68	Changes across age groups
3	`duration`	2	−0.77, +2.20	Changes: short vs. medium vs. long
4	`credit_history_critical...`	2	0.5, 0.5	Fixed binary, used in 2 branches
5	`purpose_car (new)`	2	0.5, 0.5	Fixed binary, used in 2 branches
6	`status_no checking account`	1	0.5	Fixed binary at root
7	`property_real estate`	1	0.5	Fixed binary split
8	`personal_status_sex female...`	1	0.5	Fixed binary split
9	`personal_status_sex male single`	1	0.5	Fixed binary split
10	`purpose_car (used)`	1	0.5	Fixed binary split

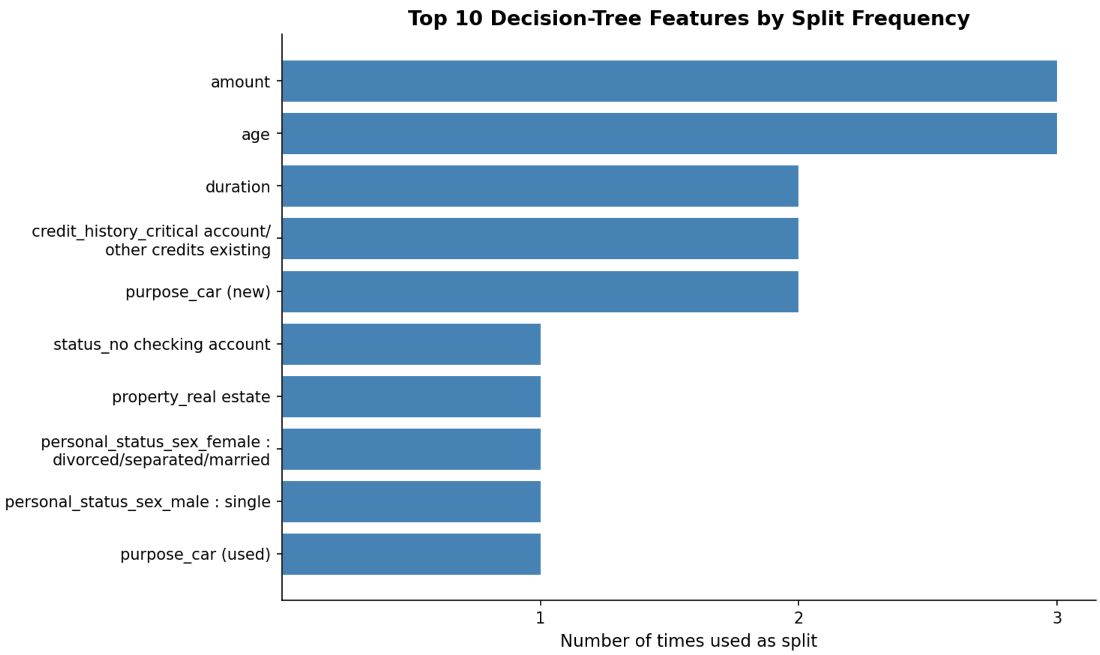

Figure 10: Top 10 Decision Tree features by number of splits (Notebook Cell 28).

Features with a changing effect (more than one split):

- `amount` (3 splits): the tree separates loan amounts into three bands (small, medium, large). The effect on credit risk is different in each band.
- `age` (3 splits): multiple age thresholds show that the relationship between age and credit risk is not simple or linear.
- `duration` (2 splits): short, medium, and very long loans are treated differently. Only very long durations (standardised value above 2.20) are a strong risk flag.

Features with a fixed effect (one split):

- `status_no checking account`, `purpose_car (used)`, `property_real estate`, and both gender-status features each appear once. They make a single yes/no decision, and that decision is the same no matter what other features the applicant has at that branch.

4.3 Cross-Model Comparison

Table 7: Agreement between LR direction and DT usage for the top features.

Feature	LR Direction	DT Splits	Note
`purpose_car (used)`	+	1	Both treat as a positive signal
`status_no checking account`	+	1	Both: dominant root feature
`purpose_car (new)`	–	2	Both treat as a negative signal
`credit_history_critical...`	+	2	Both use it; direction to check in Part 6
`other_debtors_guarantor`	+	0	LR only; not in DT
`status_...<100 DM`	–	0	LR only; not in DT
`amount`	n/a	3	DT only (numeric, 3 splits)
`duration`	n/a	2	DT only (numeric, 2 splits)

The two models agree on the direction of categorical features. Numeric features (`amount`, `duration`) are used many times in the Decision Tree but do not appear in the top-10 LR coefficients. This matches the Gini-bias discussed in Part 3 and the effect of standardisation in Logistic Regression.

Notebook Cell References

- Cell 28: DT split extraction and frequency bar chart
- Cell 29: LR coefficient table and interpretation output

Part 5: Local Explanations

We pick two individuals from the test set and explain the prediction that each model made for them. For each individual we trace the Decision Tree path from root to leaf, and we compute the Logistic Regression feature contributions (contribution = feature value × coefficient).

Individual A (dataset index 977)

Profile and Prediction

Table 8: Key features and prediction results for Individual A (Notebook Cell 31).

Field	Value
Dataset index	977
Loan duration	18 months
Loan amount	2,427 DM
Age	42 years
Checking account status	0 to 200 DM
Savings account	Unknown or no savings
Loan purpose	Other
True label	Good credit (1)
DT prediction	Good (1) — correct
LR prediction	Good (1) — correct
LR probability P(Good)	0.7720

Both models predict Good credit correctly. The LR confidence of 77.2% is above the 0.5 decision threshold but not very high, which means the individual has a mixed profile with both positive and negative signals.

Decision Tree Explanation — Path Tracing

Table 9: Step-by-step path for Individual A. Values are in standardised units (Notebook Cell 32).

Step	Feature	Value	Threshold	Condition	Direction
1	`status_no checking account`	0.0000	0.5000	$\leq$ threshold	LEFT
2	`duration`	-0.2328	-0.7679	$>$ threshold	RIGHT
3	`duration`	-0.2328	2.1959	$\leq$ threshold	LEFT
4	`amount`	-0.2740	-0.6516	$>$ threshold	RIGHT
5	`purpose_car (used)`	0.0000	0.5000	$\leq$ threshold	LEFT
Leaf node					**Good (1)**

Explanation of each step:

1. **Step 1 - Checking account.** The individual does have a checking account (value = 0, meaning the "no checking account" flag is off). The tree goes LEFT. This is the most important split in the whole tree.
2. **Steps 2 and 3 - Loan duration.** The 18-month loan gives a standardised value of -0.23. This puts it in the medium range: longer than very short loans (Step 2 goes RIGHT) but much shorter than very long loans (Step 3 goes LEFT). Medium loans are a neutral signal.

3. **Step 4 - Loan amount.** The 2,427 DM loan is above the low-amount threshold (standardised value $-0.27 > -0.65$), placing it in the mid-range. This is not a large loan, which keeps repayment risk low.
4. **Step 5 - Loan purpose.** The loan is not for a used car (value $= 0$), so the tree goes LEFT and arrives at the Good leaf.

Key finding: The most important split is Step 1 on `status_no checking account`. It is the root of the tree, and the individual's checking account status places them in the lower-risk branch right from the start.

Logistic Regression Explanation — Feature Contributions

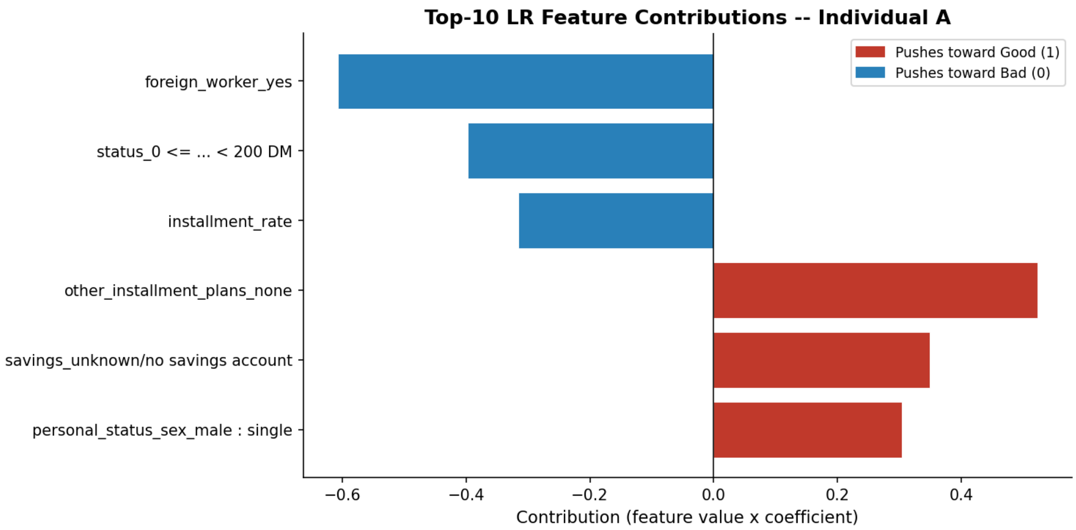

Figure 11: Top LR feature contributions for Individual A. Green bars push toward Good; red bars push toward Bad (Notebook Cell 33).

Table 10: Top positive and negative LR contributions for Individual A (Notebook Cell 33).

Direction	**Feature**	**Contribution**
Toward Good	`other_installment_plans_none`	$+0.5224$
Toward Good	`savings_unknown/no savings account`	$+0.3485$
Toward Good	`personal_status_sex_male:single`	$+0.3038$
Toward Bad	`foreign_worker_yes`	-0.6064
Toward Bad	`status_0 <= ... < 200 DM`	-0.3969
Toward Bad	`installment_rate`	-0.3159
Net result: P(Good) $= 0.7720$		Classified as Good

Features pushing toward Good:

- `other_installment_plans_none` ($+0.5224$): the strongest positive signal. The individual has no other active loans, so all repayment capacity goes to this one loan.
- `savings_unknown/no savings account` ($+0.3485$): this is counterintuitive. No savings should be a bad sign, but the model gives it a positive weight. This likely reflects a pattern in the training data and is flagged as suspicious for Part 6.

- `personal_status_sex_male:single` (+0.3038): the model gives a positive weight to being a single male. This is a gender-based signal and raises fairness concerns discussed in Part 6.

Features pushing toward Bad:

- `foreign_worker_yes` (−0.6064): the strongest negative signal. Being a foreign worker is the feature that hurts this prediction the most. Since it is linked to nationality, it is a clear fairness concern for Part 6.
- `status_0 <= ...  < 200 DM` (−0.3969): a checking account balance between 0 and 200 DM signals limited cash, which increases default risk.
- `installment_rate` (−0.3159): a higher repayment rate relative to income means the individual spends more of their money on this loan, which is a risk signal.

Final result: The positive contributions win. The biggest reason the model predicts Good is the absence of other loans. The final probability is $P(\text{Good}) = 0.7720$, which is above 0.5, so the model classifies this individual as Good credit.

Summary

Table 11: Summary of model decisions for Individual A.

	Decision Tree	**Logistic Regression**
Prediction	Good (1) - correct	Good (1) - correct
Most important feature	`status_no checking account` (root)	`foreign_worker_yes` (largest negative)
Main positive signal	Moderate duration and amount	No other active loans
Main negative signal	None (all steps led to Good)	Foreign worker status
Both agree?	Yes — both predict Good	

Both models agree that Individual A is a Good credit risk. The Decision Tree reaches this by following five conditions about checking account, loan length, loan amount, and loan purpose. The Logistic Regression reaches the same answer because having no competing loans is a strong enough positive to overcome the negative from the foreign worker status. This negative signal from `foreign_worker_yes` should be examined closely in Part 6.

Notebook Cell References

- Cell 31: Individual A raw features and predictions
- Cell 32: Decision Tree path tracing
- Cell 33: LR contributions bar chart
- Cell 34: Written summary output

Individual B (dataset index 737)

Profile and Prediction

Table 12: Key features and prediction results for Individual B (Notebook Cell 35).

Field	Value
Dataset index	737
Loan duration	18 months
Loan amount	4,380 DM
Age	35 years
Checking account status	less than 100 DM
Savings account	100 to 500 DM
Loan purpose	new car
Credit history	existing credits paid back duly till now
True label	Good credit (1)
DT prediction	Good (1) — correct
LR prediction	Bad (0) — **wrong**
LR probability P(Good)	0.4517

This individual is harder for the models to read than Individual A. The Decision Tree gets it right, but the Logistic Regression calls it Bad credit. The LR probability of 45.2% is just below the 0.5 cut-off, meaning a small shift in any one feature could flip the answer. This disagreement between the two models is worth noting.

Decision Tree Explanation — Path Tracing

Table 13: Step-by-step path for Individual B. Values are in standardised units (Notebook Cell 36).

Step	Feature	Value	Threshold	Condition	Direction
1	`status_no checking account`	0.0000	0.5000	$\leq$ threshold	LEFT
2	`duration`	−0.2623	0.1191	$\leq$ threshold	LEFT
3	`credit_history_all credits at bank paid back`	0.0000	0.5000	$\leq$ threshold	LEFT
4	`credit_history_no credits taken/all paid back`	0.0000	0.5000	$\leq$ threshold	LEFT
5	`amount`	+0.3520	−0.7166	$>$ threshold	RIGHT
Leaf node					**Good (1)**

Explanation of each step:

1. **Step 1 — Checking account.** Individual B has a checking account (the “no checking account” flag is off, value = 0), so the condition is met and the tree goes LEFT — the same first step as Individual A. This is the most important split in the whole tree.
2. **Step 2 — Loan duration.** The 18-month loan gives a standardised value of −0.26. This is below the 0.12 threshold, placing it in the shorter-duration group. The tree goes LEFT, into a different branch than Individual A took at this step.
3. **Step 3 — Credit history (all credits at this bank paid back duly).** Individual B does not have that specific flag (value = 0), so the tree goes LEFT again.
4. **Step 4 — Credit history (no credits taken / all paid back duly).** Same result: this flag is also absent (value = 0), so the tree goes LEFT once more. Steps 3 and 4 together

confirm that Individual B does not have any of the "perfect repayment" flags that would push the tree in a different direction.

5. **Step 5 — Loan amount.** The 4,380 DM loan is scaled to +0.35, which is above the −0.72 threshold. The tree goes RIGHT and arrives at a Good leaf.

Key finding: The most decisive split is again Step 1 at the root. Once the tree placed Individual B in the "has a checking account" branch, it only needed to check two credit-history flags and the loan amount to reach Good. The loan amount going RIGHT in Step 5 is what sealed the Good outcome in this branch.

Logistic Regression Explanation — Feature Contributions

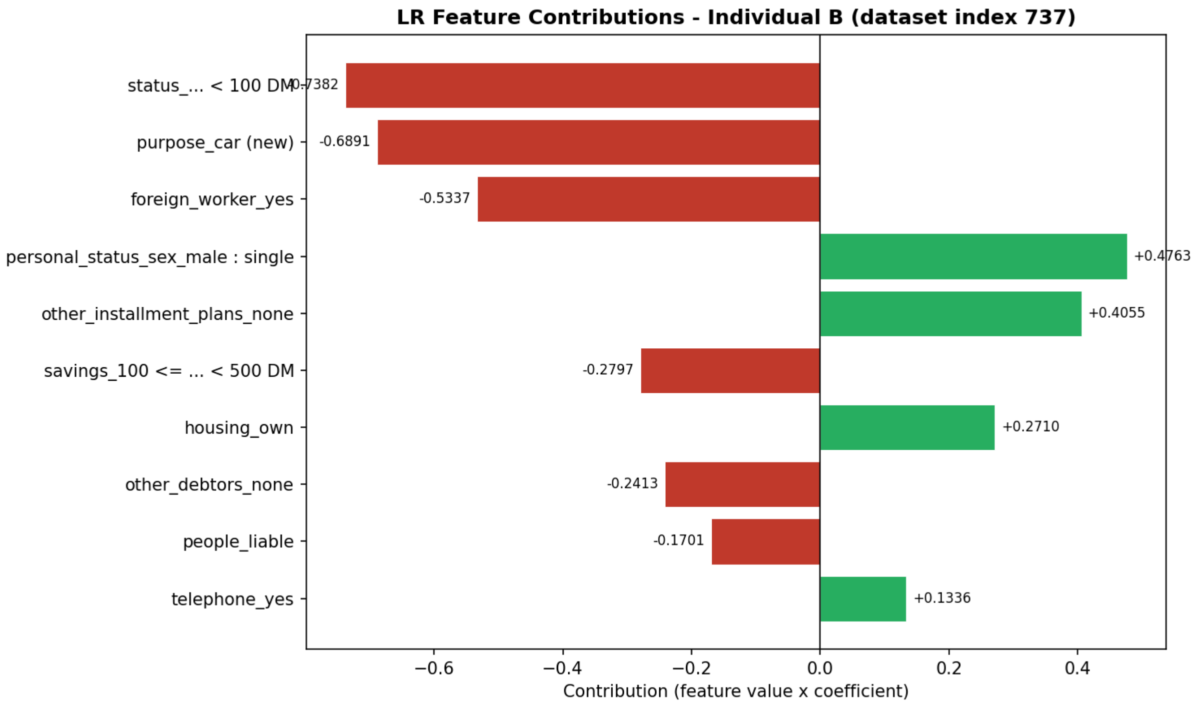

Figure 12: Top LR feature contributions for Individual B. Green bars push toward Good; red bars push toward Bad. The red bars outweigh the green ones, which is why the LR misclassifies this individual as Bad credit.

Table 14: Top positive and negative LR contributions for Individual B (Notebook Cell 37).

Direction	**Feature**	**Contribution**
Toward Good	`personal_status_sex_male:single`	+0.4763
Toward Good	`other_installment_plans_none`	+0.4055
Toward Good	`housing_own`	+0.2710
Toward Bad	`status_...<100 DM`	−0.7382
Toward Bad	`purpose_car (new)`	−0.6891
Toward Bad	`foreign_worker_yes`	−0.5337
Net result: P(Good) = 0.4517		Classified as Bad

Features pushing toward Good:

- `personal_status_sex_male:single` (+0.4763): the strongest positive signal here. Being a single male is treated by the model as a lower-risk indicator, likely because it correlates with other income or stability patterns in the training data.
- `other_installment_plans_none` (+0.4055): Individual B has no other active loans, the same positive that helped Individual A. Without competing repayments, the full income is available for this loan.
- `housing_own` (+0.2710): owning a home adds a smaller positive push. The model reads it as a sign of financial stability.

Features pushing toward Bad:

- `status_...<100 DM` (−0.7382): the strongest negative signal. Having less than 100 DM in the checking account flags a very tight cash position. This is the single biggest reason the LR leans toward Bad.
- `purpose_car (new)` (−0.6891): a new car loan is the second largest drag. New cars lose value quickly and the amounts tend to be higher than other loan types, so the model treats this purpose as a risk flag. Individual B's 4,380 DM loan is for a new car, which directly triggers this penalty.
- `foreign_worker_yes` (−0.5337): being a foreign worker is penalised here just as it was for Individual A. This is the same structural pattern flagged in Parts 4 and 6.

Final result: The negative contributions outweigh the positives. The final probability is $P(\text{Good}) = 0.4517$, which falls just below the 0.5 cut-off. This means the Logistic Regression classifies Individual B as Bad credit, even though the true label is Good. Three features — a low checking balance, a new car purpose, and foreign worker status — together pushed the prediction the wrong way.

Summary

Table 15: Summary of model decisions for Individual B.

	Decision Tree	**Logistic Regression**
Prediction	Good (1) — correct	Bad (0) — **wrong**
Most important feature	`status_no checking account` (root)	`status_...<100 DM` (largest negative)
Main positive signal	Loan amount above the low-risk threshold	No other active loans
Main negative signal	None (all steps led to Good)	Low checking balance + new car purpose
Both agree?	**No** — the models give opposite answers	

The two models reach different answers for Individual B. The Decision Tree looks at a short sequence of conditions, checking account, duration, credit history flags, and loan amount, and ends up at a Good leaf. The Logistic Regression weighs all features at once and the low checking balance combined with the new car purpose tip the total below 0.5. The fact that both models agree on Individual A but disagree here shows that the edge of the decision boundary is a sensitive place: small differences in a single feature can change the outcome depending on which model is used.

Notebook Cell References

- Cell 35: Individual B raw features and predictions
- Cell 36: Decision Tree path tracing
- Cell 37: LR contributions bar chart

- Cell 38: Written summary output

Part 6: Detect Suspicious Patterns

We now look for **questionable, unexpected, or unfair patterns** in the model. We check whether any features behave in strange ways, whether protected attributes influence decisions, and whether the model could be biased against certain groups of people.

6.1 Approval-Rate Disparities by Protected Group

We start by comparing how often each model approves applicants from different groups. A fair model should approve at roughly the same rate when the actual creditworthiness is similar. We test three protected attributes: **sex** (male vs. female), **age group** (25 and over vs. under 25), and **foreign worker status** (yes vs. no).

Table 16: Approval rates on the test set by protected group (Notebook Cell 41).

Attribute	Group	N	Actual Good %	DT Approval %	LR Approval %
Sex	Female	60	67%	85%	67%
	Male	140	73%	94%	76%
Age Group	Under 25	28	57%	86%	68%
	25 and over	172	72%	92%	74%
Foreign Worker	No	9	89%	100%	100%
	Yes	191	69%	91%	72%

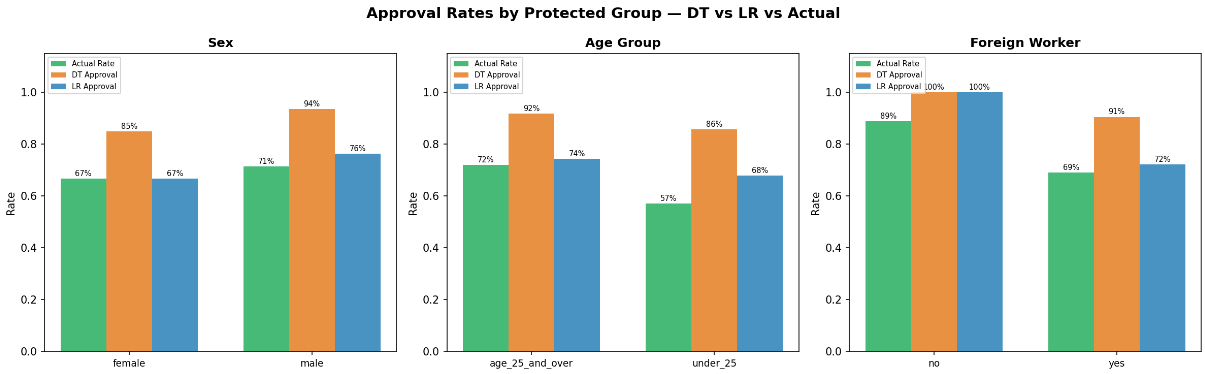

Figure 13: Approval rates by protected group for both models compared to the actual good-credit rate (Notebook Cell 41).

Key findings:

- **Sex gap.** Males are approved 9 percentage points more often by the Decision Tree (94% vs. 85%) and 10 pp more by Logistic Regression (76% vs. 67%). Part of this gap reflects the actual difference in the data (73% vs. 67%), but both models widen the gap beyond what the data alone would justify.
- **Age gap.** The Decision Tree approves under-25 applicants at 86% versus 92% for older applicants. Logistic Regression shows a similar gap (68% vs. 74%). Young applicants are penalised even more than the actual data difference (57% vs. 72%) suggests.
- **Foreign worker gap.** All 9 non-foreign-worker test applicants receive 100% approval from both models. Foreign workers get approved at 91% (DT) and only 72% (LR). The LR gap of 28 percentage points is the largest disparity we observe.

6.2 Fairness Metrics — TPR and FPR by Group

We compute two standard fairness metrics for each protected group:

- **True Positive Rate (TPR):** Among people who actually have good credit, what fraction does the model correctly approve?
- **False Positive Rate (FPR):** Among people who actually have bad credit, what fraction does the model incorrectly approve?

Table 17: Key fairness metrics by group (Notebook Cell 42).

Attribute	Group	N	DT TPR	LR TPR	DT FPR	LR FPR
Sex	Female	60	0.90	0.73	0.75	0.55
	Male	140	0.96	0.85	0.88	0.55
Age Group	Under 25	28	0.81	0.63	0.92	0.75
	25 and over	172	0.96	0.84	0.81	0.50
Foreign Worker	No	9	1.00	1.00	1.00	1.00
	Yes	191	0.94	0.80	0.83	0.54

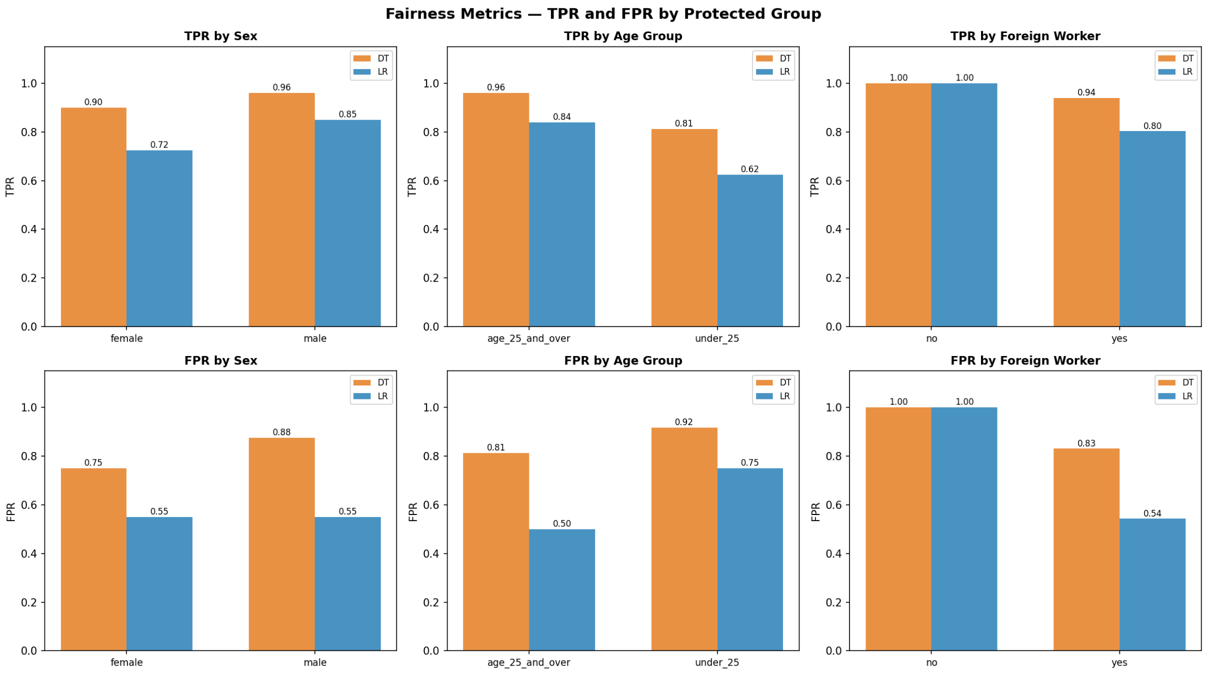

Figure 14: TPR and FPR broken down by protected group for both models (Notebook Cell 42).

Interpretation:

- **Sex TPR gap.** The LR model correctly approves 85% of creditworthy males but only 73% of creditworthy females — a 12 pp gap. This means a creditworthy woman is more likely to be wrongly rejected than a creditworthy man.
- **Age TPR gap.** The LR model has the widest age gap: TPR of 0.84 for adults versus 0.63 for under-25 applicants — a 21 pp difference. The Decision Tree also shows a gap (0.96 vs. 0.81). Young applicants who are actually creditworthy are disproportionately rejected.
- **Foreign worker TPR gap.** LR gives a perfect 1.00 TPR for domestic applicants but only 0.80 for foreign workers — a 20 pp gap. Being a foreign worker directly lowers the chance of

a correct approval.

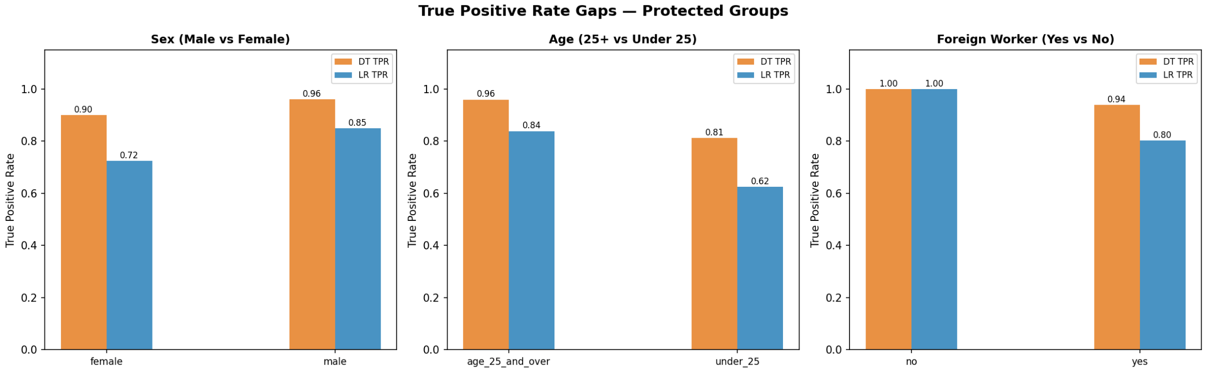

Figure 15: True Positive Rate comparison across protected groups for both models (Notebook Cell 48).

6.3 Statistical Parity Difference

Statistical Parity Difference (SPD) measures the gap in approval rates between a privileged and an unprivileged group. An SPD of zero means both groups are approved at the same rate. A positive SPD means the privileged group is favoured.

Table 18: Statistical Parity Differences (Notebook Cell 48).

Attribute	Comparison	DT SPD	LR SPD
Sex	Male vs. Female	+0.086	+0.098
Age Group	25+ vs. Under 25	+0.061	+0.066
Foreign Worker	Yes vs. No	−0.094	−0.277

Key finding: The LR model shows a foreign-worker SPD of -0.277, meaning foreign workers are approved at a rate 27.7 pp lower than domestic workers. This is a very large gap and well beyond the commonly used fairness threshold of $|SPD| \leq 0.10$.

6.4 Protected Features Used by the Models

Both models directly use features that are tied to protected attributes. We list every feature connected to sex, age, or foreign worker status, and show how much the model relies on it.

Decision Tree — Protected Feature Importance

Table 19: DT Gini importance of protected / proxy features (Notebook Cell 43).

Feature	Importance
`age`	0.0364
`personal_status_sex_female : divorced/separated/married`	0.0184
`personal_status_sex_male : single`	0.0070
Total protected importance	**0.0618 (6.2%)**

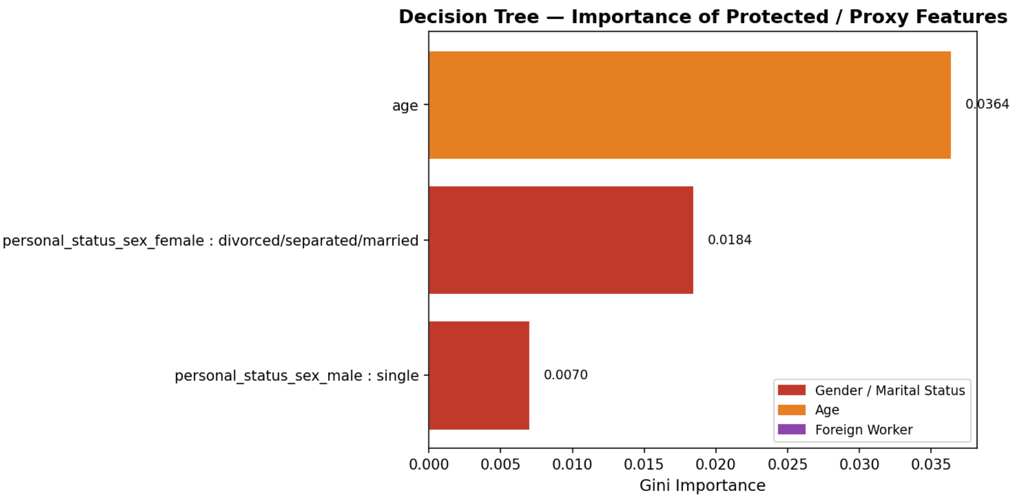

Figure 16: Decision Tree importance of protected and proxy features (Notebook Cell 43).

The Decision Tree uses `age` with 3 splits, and two gender-related features each with 1 split (see Part 4 Table 6). Together these features account for 6.2% of the total model importance. While 6.2% may seem small compared to the 36.6% contributed by checking-account status, it means that **sex and age directly change the prediction for some applicants**. A model deployed for real lending should not split on gender.

Logistic Regression — Protected Coefficients

Table 20: LR coefficients for features tied to protected attributes (Notebook Cell 44).

Feature	Coefficient	Direction
`foreign_worker_no`	+0.6124	Increases P(Good)
`foreign_worker_yes`	−0.6064	Decreases P(Good)
`personal_status_sex_male : divorced/sep.`	−0.3715	Decreases P(Good)
`personal_status_sex_male : single`	+0.3038	Increases P(Good)
`age`	+0.2568	Increases P(Good)
`personal_status_sex_male : married/wid.`	+0.1258	Increases P(Good)
`personal_status_sex_female : div./sep./mar.`	−0.0521	Decreases P(Good)
`sex_female`	−0.0521	Decreases P(Good)
`sex_male`	+0.0581	Increases P(Good)

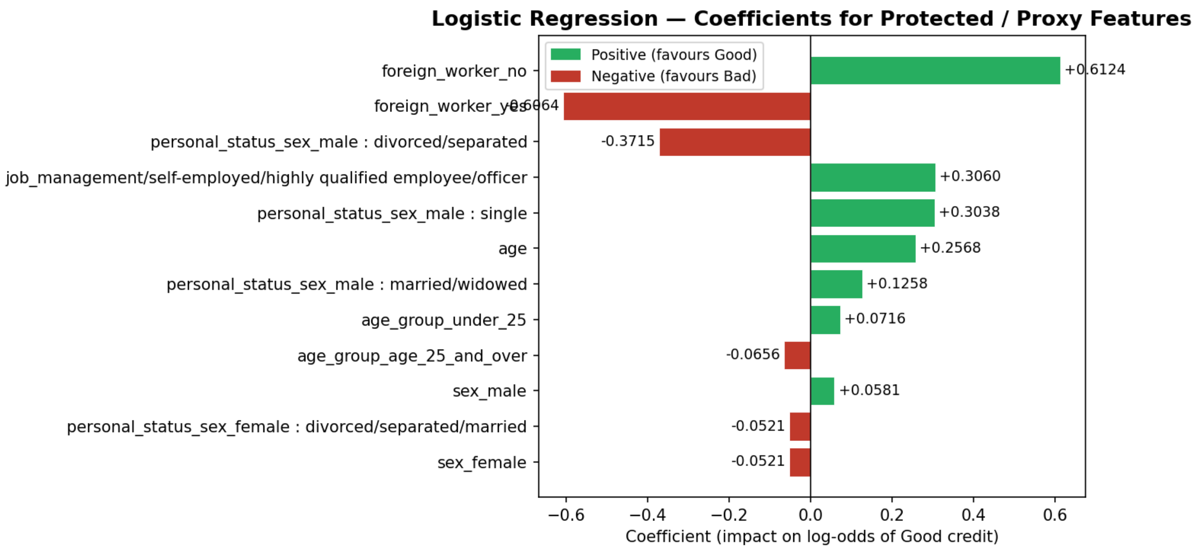

Figure 17: LR coefficients for features tied to protected attributes. Green bars favour Good credit; red bars favour Bad credit (Notebook Cell 44).

What this tells us:

- **Foreign worker penalty.** The strongest protected-attribute effect in the entire model is on foreign worker status. `foreign_worker_yes` carries a coefficient of −0.6064, meaning foreign workers start with a built-in penalty. Since 96.3% of the dataset are foreign workers, this penalty affects nearly everyone and could be considered discrimination based on nationality.
- **Gender effect.** Being female gets a small negative push (−0.0521), while being a single male gets a positive push (+0.3038). This means the model treats men and women differently, even when all other features are the same.
- **Age effect.** The coefficient for `age` is +0.2568, meaning older applicants are favoured. While age does correlate with job stability, it is a protected attribute and should not be a direct decision factor.

6.5 Counterintuitive Feature Patterns

Several features behave in the opposite direction from what a human would expect. These patterns are not necessarily wrong, but they are suspicious and deserve investigation.

Pattern 1: No Checking Account → Good Credit

`status_no checking account` is the single most important feature in both models. It has a DT importance of 0.3662 (root split) and an LR coefficient of +0.9368. In plain language: **if you have no checking account, both models are more likely to approve your loan.**

Table 21: Good-credit rate by checking-account status (full dataset).

Checking Account Status	Bad	Good	Good %
< 100 DM	135	139	50.7%
0–200 DM	105	164	60.9%
≥ 200 DM / salary assignment	14	49	77.8%
No checking account	46	348	**88.3%**

Why this is suspicious: A checking account is normally a sign of financial engagement. The fact that people *without* one are approved more often suggests a data-collection artefact. In the German banking system of the 1990s, applicants without a checking account at the lending bank may have had accounts elsewhere and were simply unknown to the lender, not necessarily unbanked. The model has learned a correlation that may not hold in a different time period or country.

Pattern 2: Critical Credit History → Good Credit

`credit_history_critical...` has an LR coefficient of +0.7832 and a DT importance of 0.0311. This means having a critical credit history **increases** the chance of being approved.

Table 22: Good-credit rate by credit-history category (full dataset).

Group	Good %	Count
Other credit histories	64.6%	707
Critical credit history	**82.9%**	293

Possible explanation: Applicants flagged as "critical" in this dataset may include people who had past problems but have since resolved them. The label may also include people with credits at other banks, which is not necessarily negative. Regardless, a model that rewards "critical" history is hard to explain to a customer or regulator and could mask risky applicants.

Pattern 3: No Savings → Good Credit (LR only)

`savings_unknown/no savings account` has an LR coefficient of +0.3485, meaning that having no savings pushes the prediction toward Good credit. This is the opposite of what one would expect: savings are a financial safety net, and their absence should increase risk. This feature has zero importance in the Decision Tree, so the DT does not rely on it.

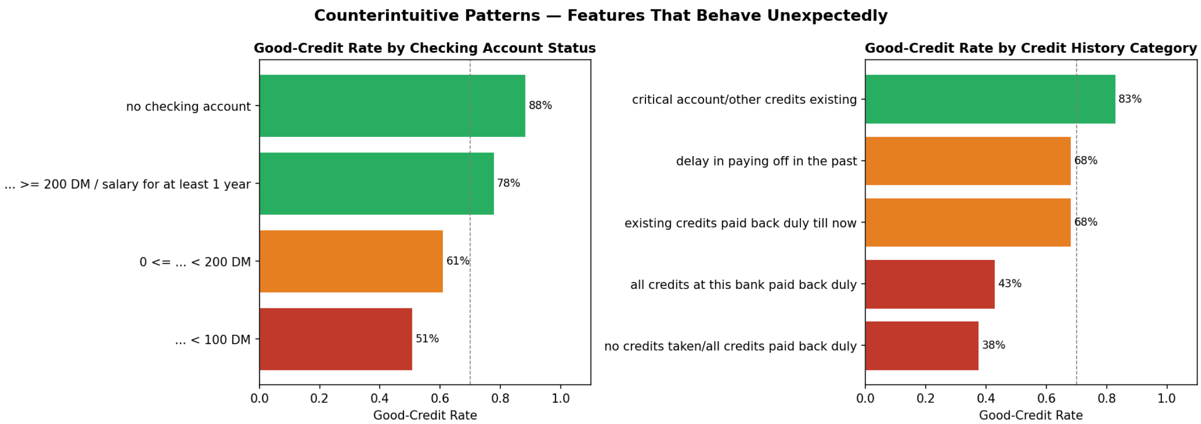

Figure 18: Good-credit rate by checking-account status (left) and credit-history category (right). The highest good-credit rates correspond to the categories whose names sound the most risky (Notebook Cell 45).

6.6 Proxy Correlation Analysis

Even if a model does not use a protected attribute directly, it can still be biased through **proxy features** — features that correlate with protected attributes. We compute the correlation between protected attributes and the key features used by the model.

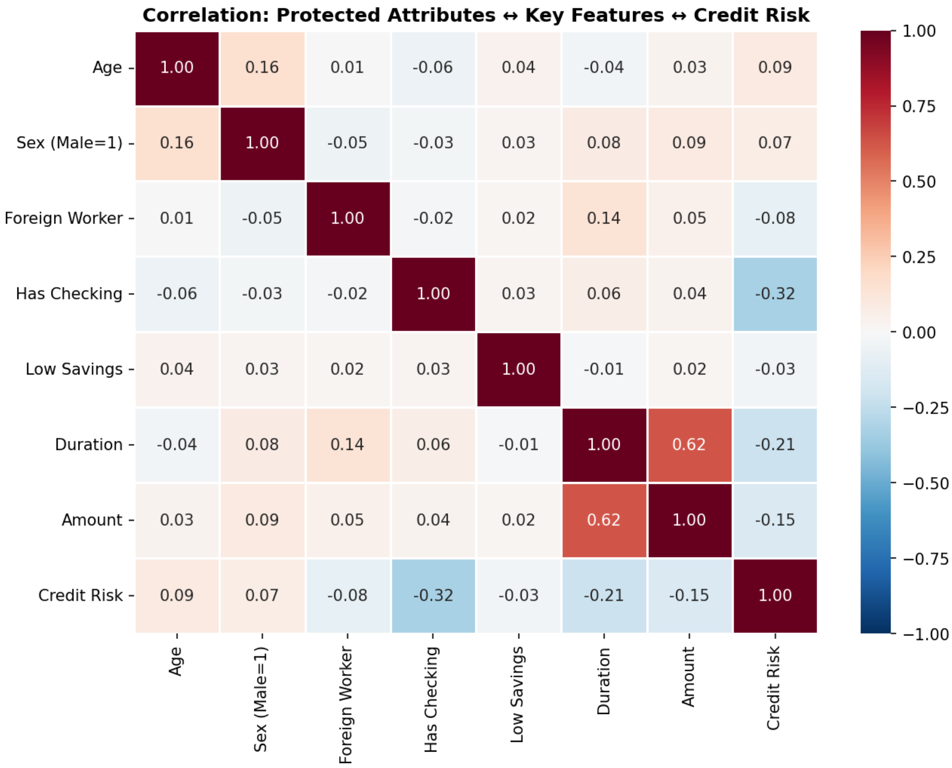

Figure 19: Pearson correlation between protected attributes, key features, and credit risk (Notebook Cell 46).

Notable correlations:

- `has_checking` and `credit_risk`: $r = -0.322$. This is the strongest single-feature correlation with the target, confirming why both models rely so heavily on checking-account status.
- `foreign_worker` and `duration`: $r = +0.138$. Foreign workers tend to take out longer loans. Since longer loans increase risk, foreign-worker status acts as an indirect risk signal through duration, even if foreign-worker status were removed, the model could still penalise this group through the duration feature.
- `sex` and `age`: $r = +0.162$. Males in the dataset are slightly older. Since age is used by both models and favours older applicants, the age feature indirectly favours males over females.
- Most correlations between protected attributes and features are weak ($|r| < 0.1$), which means the main source of bias is the **direct use** of protected features (foreign worker, sex, age), not hidden proxies.

6.7 Age Distribution — Approved vs. Rejected

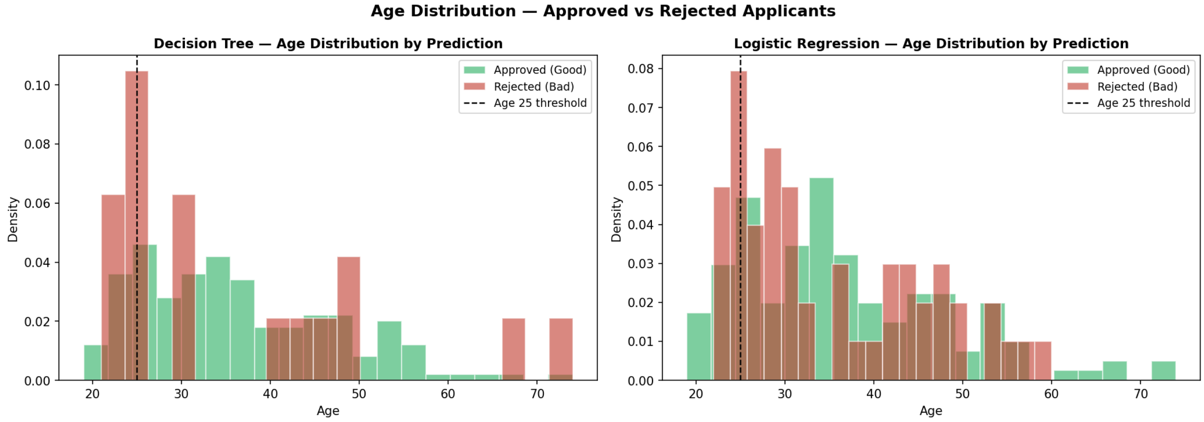

Figure 20: Age distribution of approved vs. rejected applicants for both models. The dashed line marks age 25, the boundary used for the protected age-group attribute (Notebook Cell 47).

In the Decision Tree, the mean age of approved and rejected applicants is almost identical (36.2 years in both cases), showing that the DT does not heavily penalise young applicants on average. In the Logistic Regression, approved applicants are slightly older on average (36.6 vs. 35.1 years), and fewer under-25 applicants are approved (19 out of 28 vs. 24 out of 28 in the DT). This aligns with the positive `age` coefficient (+0.2568) in the LR model.

6.8 Comparison with Module 1 Fairness Findings

In Module 1, we audited a baseline Logistic Regression model for fairness across Gender and Age. Now in Module 2, we are using a Decision Tree. We need to check if changing the model fixed the unfairness, or if the bias is still there.

Here is a simple comparison between the fairness gaps we found in Module 1 and our new Decision Tree model in Module 2.

Table 23: Comparison of Fairness Gaps: Module 1 (LR) vs. Module 2 (Decision Tree).

Protected Attribute	Fairness Metric	Module 1 (LR Baseline)	Module 2 (Decision Tree)
Gender (Male vs. Female)	Approval Rate Gap (Demographic Parity)	0.0976	0.0860
	Detection Gap (TPR Gap)	0.1250	0.0600
	Wrong Approval Gap (FPR Gap)	0.0000	0.1300
Age (25+ vs. Under 25)	Approval Rate Gap (Demographic Parity)	0.0656	0.0610
	Detection Gap (TPR Gap)	0.2137	0.1500
	Wrong Approval Gap (FPR Gap)	0.2500	0.1100

What did we learn from this comparison in simple words?

- **Did the new model fix the Age bias?** It helped, but did not completely fix it. In Module 1, the TPR gap (meaning how many actually creditworthy young people are missed) was very high at 0.2137. The Decision Tree lowered this gap to 0.1500. This is better, but a gap of 15% is still considered unfair. The FPR gap (wrongly approving bad applicants) also dropped from 0.2500 down to 0.1100.

- **Did the new model fix the Gender bias?** Not really; it just shifted the problem. In Module 1, males and females had the exact same FPR (gap of 0.0000). But now, in the Decision Tree, the FPR gap jumped to 0.1300, meaning the model is now much more likely to make a mistake and approve an unqualified male than an unqualified female. While the TPR gap improved (from 0.1250 down to 0.0600), the unequal treatment is still there.
- **Foreign Workers (New in Module 2):** We did not analyze foreign workers in Module 1, but our Module 2 analysis found this to be the biggest bias of all. The Logistic Regression model heavily penalizes foreign workers purely for their nationality (see Section 6.4.2).

Conclusion on Bias: Switching to a Decision Tree does not magically fix historical bias. The data itself holds biases against females, young people, and foreign workers. Because the Decision Tree learned from the same biased data, it continues to show unfair patterns.

6.9 Summary of Suspicious Patterns

Table 24: Summary of all suspicious patterns found in Part 6.

Finding	**Details**	**Risk**
Foreign worker penalty	LR coef = −0.61; SPD = −0.28; 96% of dataset affected	Nationality-based discrimination
Gender in model	DT splits on sex (imp = 0.025); LR gives females negative weight	Violates equal treatment based on sex
Age as split feature	DT uses age in 3 splits; LR coef = +0.26	Age discrimination against younger applicants
No checking → Good	88% good-credit rate; root split of DT; LR coef = +0.94	Data artefact, may not hold outside this dataset
Critical history → Good	83% good-credit rate; LR coef = +0.78	Counterintuitive; masks risky applicants
No savings → Good	LR coef = +0.35	Counterintuitive; opposite of financial logic

Overall verdict. The model has learned several patterns that are **ethically questionable**. The most serious issue is the direct use of `foreign_worker_yes` in Logistic Regression, which penalises nearly every applicant and produces a Statistical Parity Difference of −0.277. Gender features appear in both models, and age is used as a split feature with 3 thresholds in the Decision Tree. On top of that, three features behave in the opposite direction from what common financial sense would predict, making the model harder to trust and harder to explain to customers.

Part 7: Design Recommendation

We now step back and look at the full picture. We answer three questions: would we put this model into a real bank system? What could go wrong? And what would we change before that could happen?

7.1 Would We Use This Model in a Real Lending System?

The short answer is **no, not in its current form**.

Both models can explain their decisions in plain language. A bank officer can follow the five splits that led to a Decision Tree verdict, or read the list of feature contributions for a Logistic Regression score. That transparency is genuinely useful, and it is the main reason these two models were chosen over less explainable alternatives.

However, the analysis in Part 6 found three problems that would make deployment legally and ethically unjustifiable:

1. **Nationality is a direct input.** The Logistic Regression gives `foreign_worker_yes` a coefficient of -0.6064 and `foreign_worker_no` a coefficient of $+0.6124$ (Notebook Cell 44). The gap between these two values is 1.22 , larger than the gap for any other binary pair in the model. The result is a 27.7-point approval-rate difference between non-foreign and foreign workers (SPD = -0.277, Notebook Cell 48). Deciding who gets a loan partly based on nationality is prohibited under anti-discrimination law in most countries.
2. **Sex and age are direct inputs.** The Decision Tree splits on two gender-related columns and uses `age` at three thresholds, together accounting for 6.2% of the tree's total Gini importance (Notebook Cell 43). The Logistic Regression gives being female a negative push (-0.0521) and being a single male a positive push ($+0.3038$) (Notebook Cell 44). Applicants under 25 are correctly identified as creditworthy at a TPR of only 0.63 in the LR, compared to 0.84 for those aged 25 and over (Notebook Cell 42). A 21-point gap in correctly approving young adults is a direct fairness failure.
3. **Three features work backwards.** Having no checking account, a critical credit history, and no savings all *increase* the probability of being approved (Notebook Cell 45). A bank cannot explain to a regulator, or to a customer, why the absence of a checking account is treated as a good sign. If the pattern shifts in a new population of applicants, the model will produce large unexpected losses.

7.2 What Are the Risks?

The risks fall into two groups: fairness risks that could harm people, and reliability risks that could harm the bank.

Fairness Risks

- **Foreign worker applicants are systematically under-approved.** 191 out of 200 test applicants (95.5%) are foreign workers (Notebook Cell 41). The LR approves them at only 72.3%, compared to 100% for the 9 non-foreign workers. Because the feature `foreign_worker_yes` is baked directly into the model weights, this gap cannot be fixed by adjusting the decision threshold alone, the feature itself would need to be removed.
- **Young applicants who should be approved are often rejected.** The LR has a TPR of only 0.625 for applicants under 25, versus 0.839 for those 25 and over (Notebook Cell 42). This means that roughly one in three creditworthy young applicants is wrongly turned

down. Since `age` is used directly as a split feature in the Decision Tree (3 splits, importance = 0.0364) and as a positive coefficient in the LR (+0.2568), the discrimination is structural.

- **Women are more likely to be wrongly rejected than men.** The LR correctly approves 85% of creditworthy men but only 73% of creditworthy women, a 12-point gap in TPR (Notebook Cell 42). The Decision Tree also approves men at a higher rate (93.6% vs. 85.0%, SPD = +0.086, Notebook Cell 48). These gaps are consistent across both models, which means the bias comes from the data, not from the choice of model.

Reliability Risks

- **The three counterintuitive patterns may not hold on new data.** The no-checking-account group has an 88.3% good-credit rate in the training data (Notebook Cell 45). If this pattern is specific to German banking in the 1990s, a model deployed today could incorrectly approve a large wave of bad-credit applicants simply because they lack a checking account at the lending bank.
- **The models over-approve relative to the true rate.** The Decision Tree approves 90.5% of the overall test set while the true good-credit rate is 71.0%. The False Positive Rate reaches 0.875 for male applicants and 0.917 for applicants under 25 (Notebook Cell 42). This means the bank would expect to grant loans to roughly one in four people who will not repay, which is a significant financial exposure.
- **The non-foreign-worker group is too small to measure fairly.** There are only 9 non-foreign workers in the test set. Every fairness number reported for that group (100% approval, TPR 1.00, FPR 1.00) is based on 9 data points and is therefore unreliable. Any real deployment decision would need a much larger sample of that group before any claim about equal treatment could be trusted.
- **The dataset is over 30 years old.** The German Credit Dataset was collected in the early 1990s. Loan amounts, interest rates, employment patterns, and borrower behaviour have all changed substantially since then. A model trained on this data is likely to produce decisions that reflect a past reality rather than current risk.

7.3 What Would We Change?

Four changes are needed before this system could be considered for any real lending decision.

1. **Remove protected attributes from the feature set.** The columns `foreign_worker`, `sex`, `age_group`, and all `personal_status_sex` columns should be dropped before training. This directly removes the nationality gap (SPD = −0.277), the gender effect (LR coef range from −0.3715 to +0.3038), and the direct age discrimination (DT uses age at 3 thresholds, LR coef = +0.2568). The models would then need to be re-trained on the reduced feature set, and all performance and fairness metrics re-measured. Note that removing these features will not completely eliminate bias, the proxy correlation in Notebook Cell 46 shows that `duration` is correlated with foreign worker status ($r = +0.138$), so some indirect effect may remain.
2. **Investigate and resolve the three counterintuitive patterns with a domain expert.** Before any deployment, a credit analyst should examine why no checking account (88.3% good-credit rate), critical credit history (82.9% good-credit rate), and no savings all push toward approval (Notebook Cell 45). If these reflect a genuine and stable economic behaviour, the features can stay. If they are artefacts of how the 1994 dataset was recorded, they should be removed or the model should be constrained so it cannot use them. A model whose key

feature works against common sense needs a clear written justification before a regulator would accept it.

3. **Apply fairness-aware constraints during re-training.** After removing protected features, fairness gaps may still exist through proxy features. The target should be to bring every group's TPR gap below 5 percentage points and every SPD below ± 0.10. Practical options include reweighting under-represented groups in the training data (e.g. upweighting female and under-25 applicants, whose true good-credit rates are 66.7% and 57.1% respectively, much lower than the overall 71%), or applying a post-processing step that adjusts the decision threshold separately for each group until the TPR gaps close.

4. **Replace the dataset with recent, balanced data.** A model used for real lending should be trained on the bank's own recent loan book, not a 30-year-old research dataset. The new data should be checked to ensure that protected groups are represented in sufficient numbers, at least large enough that fairness metrics are statistically meaningful. The 9 non-foreign-worker applicants in the current test set are a clear example of a group that is too small to audit properly.

Notebook Cell References

- Cell 41: Approval rates by protected group (basis for Section 7.1)
- Cell 42: TPR and FPR gaps (basis for fairness risk assessment)
- Cell 43: DT protected feature importance — age 6.2% of total
- Cell 44: LR coefficients for protected features — foreign worker gap 1.22
- Cell 45: Counterintuitive patterns — no checking account 88.3%, critical history 82.9%
- Cell 46: Proxy correlation — duration and foreign worker $r = +0.138$
- Cell 48: Statistical Parity Differences — SPD $= -0.277$ for foreign workers